THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 14: **Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Take $\pi = \frac{22}{7}$. The curved surface area of a cylinder of radius 3.5 cm and height 11 cm is: 1

(a) 217 cm^2 **(b)** 267 cm^2 **(c)** 292 cm^2 **(d)** 242 cm^2

2. Take $\pi = \frac{22}{7}$. The total surface area of a cylinder of radius 3.5 cm and height 27 cm is: 1

(a) 603 cm^2 **(b)** 807 cm^2 **(c)** 739 cm^2 **(d)** 671 cm^2

3. Take $\pi = \frac{22}{7}$. The curved surface area of a cone of radius 14 cm and slant height 50 cm is: 1

(a) 2421 cm^2 **(b)** 1979 cm^2 **(c)** 2642 cm^2 **(d)** 2200 cm^2

4. Take $\pi = \frac{22}{7}$. The volume of a cone of radius 10.5 cm and height 14 cm is: 1

(a) 1617 cm^3 **(b)** 1942 cm^3 **(c)** 1779 cm^3 **(d)** 1455 cm^3

5. Take $\pi = \frac{22}{7}$. The surface area of a sphere of radius 35 cm is: 1

(a) 18482 cm^2 **(b)** 15400 cm^2 **(c)** 16941 cm^2 **(d)** 13859 cm^2

6. Take $\pi = \frac{22}{7}$. The volume of a sphere of radius 31.5 cm is: 1

(a) 157174 cm^3 **(b)** 144075 cm^3 **(c)** 117879 cm^3 **(d)** 130977 cm^3

7. Take $\pi = \frac{22}{7}$. The curved surface area of a hemisphere of radius 3.5 cm is: 1

(a) 85 cm^2 **(b)** 69 cm^2 **(c)** 94 cm^2 **(d)** 77 cm^2

8. The volume of a pyramid with base area B and height h is: 1

(a) $\frac{1}{3}Bh$ **(b)** Bh **(c)** $\frac{1}{2}Bh$ **(d)** $\frac{2}{3}Bh$

9. Take $\pi = \frac{22}{7}$. The curved surface area of a cylinder of radius 28 cm and height 27 cm is: 1

(a) 5704 cm^2 **(b)** 5228 cm^2 **(c)** 4752 cm^2 **(d)** 4276 cm^2

10. Take $\pi = \frac{22}{7}$. The total surface area of a cylinder of radius 10.5 cm and height 18 cm is: 1

(a) 1881 cm^2 **(b)** 1692 cm^2 **(c)** 2259 cm^2 **(d)** 2070 cm^2

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Take $\pi = \frac{22}{7}$. A cone has radius 7 cm and height 24 cm. Find its slant height, curved surface area and volume. 2

12. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area of a solid hemisphere of radius 28 cm. 2

13. A pyramid has a square base of side 9 cm and height 36 cm. Find its volume. 2

14. Take $\pi = \frac{22}{7}$. A cone has radius 42 cm and height 56 cm. Find its slant height, curved surface area and volume. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Take $\pi = \frac{22}{7}$. A cone has radius 14 cm and height 48 cm. Find its slant height, curved surface area and volume. 3

16. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a solid hemisphere of radius 52.5 cm. 3

17. Take $\pi = \frac{22}{7}$. A cylindrical pillar has radius 7 cm and height 35 cm. Find the cost of painting its curved surface at ₹10 per 1000 cm^2. 3

OR

Take $\pi = \frac{22}{7}$. A cone has radius 10.5 cm and height 14 cm. Find its slant height, curved surface area and volume. 3

18. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a solid hemisphere of radius 63 cm. 3

OR

Take $\pi = \frac{22}{7}$. A cylindrical pillar has radius 14 cm and height 25 cm. Find the cost of painting its curved surface at ₹10 per 1000 cm^2. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A rectangular sheet 44 cm long and 25 cm wide is rolled so that its length forms the circumference of the base of a cylinder of height 25 cm. Find the radius, the curved surface area and the volume (Take $\pi = \frac{22}{7}$). 5

OR

A solid metal sphere of radius 8 cm is melted and recast into small solid spheres of radius 2 cm. Find the number of small spheres and compare the total surface area of the small spheres with the surface area of the big sphere. 5

20. A rectangular sheet 44 cm long and 20 cm wide is rolled so that its length forms the circumference of the base of a cylinder of height 20 cm. Find the radius, the curved surface area and the volume (Take $\pi = \frac{22}{7}$). 5

OR

A solid metal sphere of radius 9 cm is melted and recast into small solid spheres of radius 3 cm. Find the number of small spheres and compare the total surface area of the small spheres with the surface area of the big sphere. 5

END OF QUESTION PAPER — Total: 40 Marks